Common Terms/Definitions/Explanations

- **UART**
  - UART Baud rate (bits per second). Rate in which UART transmitter transmit information (START, STOP, DATA, PARITY).
  - Data rate = Data transfer rate = Effective Data transfer rate = rate in which 'data' (e.g., excluding START, STOP, and PARITY) is transferred.

- **Memory units**
  - 1KByte = 2^10 Bytes
  - 1MByte = 2^20Bytes
  - 1GByte = 2^30 Bytes
  - 1TByte = 2^40 Bytes
  - 1Kbps = 1,000bps (bits per sec)
  - 1Mbps = 1,000,000bps
  - 1KB/s = (2^10)Bytes/sec = 1,024Bytes/sec

- **Computer Memory (general)**
  - Internal memory is a group of memories that are 'internal' to the processor, it is part of the processor, together with the registers, cache, TLB, etc. This is versus external memory which is separate from the processor.
  - In this part of the course, System Memory = Main Memory = Physical Memory; They are used to stress different aspects of properties/ functionalities for a same memory class throughout this course. In many situations, they refer to External Memory, since Internal Memory is small in size and does not access by general system buses.
  - "Memory types" refers to SRAM, DRAM, NOR Flash, NAND Flash, HDD, etc.
  - "System memory" and "Storage Memory" are functional partition of a computer memory system. System memory stores runtime code/data and processor can execute code directly from it. Storage memory 'stores' all code/data and processor doesn't execute code directly from it.

- **Cache**
  - Cache block = cache line = one 'row' of cache in the cache diagram.
  - Offset (Cache) = the relative position of the target byte from the start of the cache block.
  - Effective Access Time equation:

    [A] EAT = H x $Access_C$ + (1-H) x *Miss Penalty.*

    [B] EAT = H x $Access_C$ + (1-H) x ($Access_C$ + $Access_{MM}$).

    - Expression [A] is the generic form that applies to all systems if you are able to derive the "Miss Penalty".
    - Expression [B] is a simplified form under the assumptions stated in the lecture notes. It applies only to <u>READ</u> operations as write

operations will be more complicated (with combination of write-through/back, write-allocate/no-allocate policies).
- These expressions are for "steady state" performance, i.e. when you transfer a huge amount of data (GBytes).
- For small transfers of a few to tens of bytes, you need to analyse the operation of the cache, e.g. if a 10-byte array is not in the cache initially, then following timing is needed to read the 10 bytes:
  - First byte (cache miss) => time taken = one cache read + one MM read.
  - Rest of the 9 bytes (cache hit) => time taken = 9 cache reads. (Assume: 10 bytes are within a same cache block that has been transferred to cache together with the 1st byte (when the cache miss happens)).

- **Virtual Memory (VM)**
  - Offset (in VM) = the relative position of the target byte from the start of the virtual page or physical frame.
  - 'PAGE' field in VM refers to the Page Number of the Virtual Page which contains the target byte.
  - 'FRAME' field in VM refers to the Frame Number of the Physical Frame which contains the target byte.
  - Virtual Memory address = the address that program code uses. Virtual memory address is a logical entity, and it is "not real". The OS will translate the virtual address to a physical address that is used to locate the required code/data in the physical memory (this is the 'real' memory where code/data is stored).
  - Physical Memory address = address where code/data is stored in the Physical/Main/System memory during runtime.
  - TLB is a subset of PAGE TABLE; PAGE TABLE resides in Main Memory and contains entries for all virtual pages (valid and invalid).

- **Difference between Cache and TLB**
  - Cache stores frequently used code/data.
  - TLB (a page table cache) stores frequently used VirtualPage to PhysicalFrame translation information.

- **Pipeline**
  - For a 4-stage pipeline: F, D, E and S.
    - Fetch: Instruction (machine code) is fetched into the pipeline
    - Decode: The machine code is decoded to know which instruction it correspond to.
    - Execute: The instruction is executed, e.g. ADD, SUB operations.
    - Store: The result of the instruction execution is stored to the corresponding destination register or memory.

- You can assume that the status flag is updated at the E-stage of a pipeline unless otherwise stated.
- "Delay slots" are the slots after a Branch instruction that may be flushed if the branch is TRUE.
  - There are two delay slots if the branch decision and target address are resolved at the E-Stage.
  - If the branch decision and target address are resolved at the D-Stage, there is only one delay slot.
- An instruction located in the delay slot is known as a delay slot instruction.
- For pipeline related questions, all the 'ARM' instructions presented will affect the status flag, i.e. ADD = ADDS, MOV = MOVS, etc.
- Explanation of items (indicated with bullets below) in pipeline questions.

  e.g., consider a processor with 4 pipeline stages: F, D, E and S.
  Assume that (unless specified otherwise):

  - Branch target address is calculated at the execute stage

    Meaning: Branch decision and target address are resolved at the E-Stage.
  - Instruction length for every instruction is one word long

    To simplify pipeline analysis; implies that each instruction only occupies one row of the pipeline diagram.
  - Each pipeline stage takes 1 cycle to complete

    In pipeline processor, one pipeline stage takes 1 CPU cycle, so the first instruction needs 4 cycles to travel through the whole pipeline.
  - Delayed branching is <u>disabled</u>.

    It means that instructions in the delay slot will be discarded if the branch is TRUE.
  - This processor is <u>not</u> an ARM processor. All instructions will update the relevant status flags.
    - Only consider the function of the 'ARM' assembly instructions you see in the question. All instructions in the question are one word long and all will affect the status bits (where applicable).

- **Comms and UI**
  - "Channel Bandwidth" refers to the frequency range used for transmission. Note that '2.4 GHz' is not a single point frequency, and it encompasses a range of frequency from around 2.4Ghz to 2.48 Ghz. This range is further divided in 'channels' of around 22Mhz each in bandwidth for Wifi Standard, e.g. Channel 6 is a RF channel with 22-Mhz bandwidth channel, centered at 2.437Ghz.
  - Typically, higher bandwidth means
    - Higher possible data transfer rate if there are no interference.

  - Higher chance of being interfered by other RF transmitters.
  - You can visualise bandwidth as the diameter of a water pipe: with a larger diameter, more water flow but higher chance to encounter interference. (If you consider the signal to be a square wave, it is clear that higher bandwidths allow higher frequency square waves to be transmitted.)
- "Royalty" is amount of fees paid to use a particular service or product. In this case, the RF frequency band.
- "Attenuation" refers to the drop in RF signal amplitude as RF wave travels across the medium e.g. air.
- "Interference" occurs when two RF waves meet and cause some constructive or destructive interference to each other. It occurs even if the two waves are carrying signals from different standards, e.g. Wifi and Bluetooth.